Solución tarea 2-6

1. Creamos el directorio “entrada”

Lo creamos con el comando mkdir, el comando ls es para comprobar que se creó correctamente.

```
Applications  Places  System                                   Sun Mar 8, 3:31 AM  cloudera
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ mkdir entrada
[cloudera@quickstart ~]$ ls
cloudera-manager  cm_api.py  Desktop  Documents  Downloads  eclipse  enterprise-deployment.json  entrada  express-deployment.json  kerberos  lib  Music  parcels  Pictures  Public  Templates  Videos  workspace
[cloudera@quickstart ~]$
```

2. Movemos el archivo novela.txt al directorio entrada

Con el comando mv movemos el archivo a la ruta que queremos

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ mv novela.txt entrada/
[cloudera@quickstart ~]$ ls entrada
novela.txt
[cloudera@quickstart ~]$
```

3. Cargamos las librerías de Hadoop

```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ export HADOOP_CLASSPATH=$(hadoop classpath)
[cloudera@quickstart ~]$
```

4. Compilamos el programa WordCount.java

```
cloudera@quickstart ~]$ javac -classpath $HADOOP_CLASSPATH -d . WordCountMapReduce.java
cloudera@quickstart ~]$ find . -name "*.class"
/WordCountMapReduce.class
/WordCountMapReduce$Reduce.class
/workspace/training/bin/StubReducer.class
/workspace/training/bin/StubMapper.class
/workspace/training/bin/StubDriver.class
/workspace/training/bin/StubTest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactInfoFilter.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/UniqueArtifactFilterPostprocessor.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/CleaningEncoder.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/GroupedSearchRequest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactContext$ModelReader.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/AbstractMultiArtifactInfoFilter.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/IteratorSearchRequest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/GroupedSearchResponse.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultSearchEngine.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/FlatSearchRequest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/IteratorSearchResponse$1.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/AndMultiArtifactInfoFilter.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/MatchHighlight.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactContextProducer.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactScanningListener.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/FlatSearchResponse.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultScanner.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/MatchHighlightRequest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/OneLineFragmenter.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/UniqueGAArtifactFilterPostprocessor.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultArtifactContextProducer.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/SearchType.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/Field.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultQueryCreator.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/AbstractSearchResponse.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ScanningResult.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/NEXUS.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactAvailablility.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ArtifactInfoRecord.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultScanner$1.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/IndexerField.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/DefaultScannerListener.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/ScanningRequest.class
/eclipse/plugins/org.eclipse.m2e.maven.indexer_1.5.1.20150109-1819/org/apache/maven/index/IteratorSearchResponse.class
```

5. Creamos el archivo JAR

Empaquetamos los .class para ejecutarlos con Hadoop

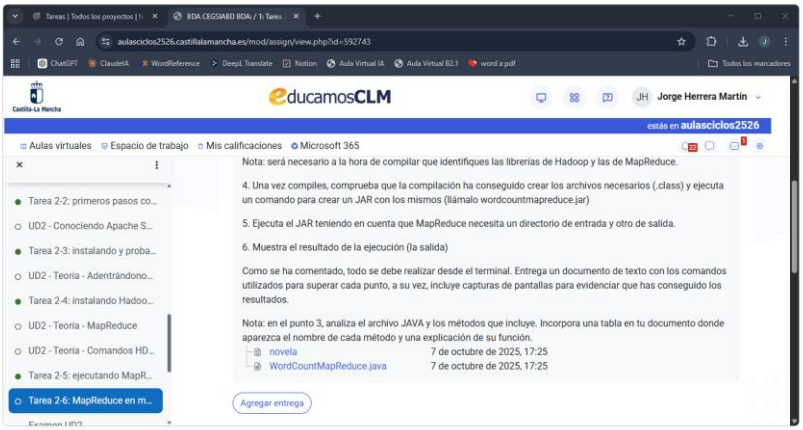
```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ jar -cvf wordcountmapreduce.jar *.class
added manifest
adding: WordCountMapReduce.class(in = 1991) (out= 986)(deflated 50%)
adding: WordCountMapReduce$Map.class(in = 2299) (out= 1036)(deflated 54%)
adding: WordCountMapReduce$Reduce.class(in = 1654) (out= 689)(deflated 58%)
[cloudera@quickstart ~]$ ls
cloudera-manager  Desktop    Downloads  enterprise-deployment.json  express-deployment.json  lib    parcels   Public    Videos                    wordcountmapreduce.jar  WordCountMapReduce$Map.class    workspace
cm_api.py         Documents  eclipse    entrada                     kerberos                 Music  Pictures  Templates WordCountMapReduce.class  WordCountMapReduce.java WordCountMapReduce$Reduce.class
[cloudera@quickstart ~]$
```

6. Subimos los archivos a HDFS

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hdfs dfs -mkdir entrada
[cloudera@quickstart ~]$ hdfs dfs -put entrada/novela.txt entrada/
[cloudera@quickstart ~]$ hdfs dfs -ls entrada
Found 1 items
-rw-r--r--   1 cloudera cloudera     131797 2026-03-08 04:00 entrada/novela.txt
[cloudera@quickstart ~]$
```

7. Ejecutamos el WordCount

Comando: hadoop jar wordcountmapreduce.jar WordCountMapReduce entrada salida

Para ver la salida utilizo el comando: hdfs dfs -cat salida/part-r-00000

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
vistas  1
visto   5
vistoso 1
visual  2
vital   2
vivencias       1
vivido  1
vivienda        1
viviendas       1
viviendo        1
viviera 1
vivieran        1
vivir   5
vivo    2
vivos   1
vocabulario     1
vocales 2
voces   1
volar   2
volatilizado    1
voluntad        1
volver  16
volverse        1
volviendo       5
volviera        1
vomitar 2
voraz   1
vosotros        1
voy     2
voz     19
vuelo   1
vuelos  1
vuelta  4
vueltas 1
vuelto  2
vuelve  1
web     1
y       413
ya      41
yang    1
yema    1
yemas   1
yendo   2
yente   1
yin     1
yo      12
zapato  1
zarrio  1
zona    5
[cloudera@quickstart ~]$
```